

1 Starting equations (partial differential equations)

1.1 Energy equation

1.1.1 CRB

CRB: Cushman-Roisin and Beckers, 2011

$$\rho C_v \frac{dT}{dt} + p \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) = k_T \nabla^2 T \quad (1)$$

I refer to Schubert et al., 2001 for the conversion into the constant-pressure heat capacity variant of this equation.

1.1.2 Jarvis & McKenzie, 1980

jamc80 Jarvis and McKenzie, 1980:

$$\rho C_p \left(\frac{\partial T}{\partial t} - \frac{\alpha T}{\rho C_p} \frac{\partial P}{\partial t} + \mathbf{u} \cdot [\nabla T - (\nabla T)_s] \right) = \nabla \cdot (K \nabla T) + H + \tau_{ij} \frac{\partial u_i}{\partial x_j} \quad (2)$$

1.1.3 Gassmüller et al., 2020

Gassmüller: Gassmüller et al., 2020

$$\rho C_p \left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) - \nabla \cdot (k \nabla T) = \rho H + \tau : \dot{\epsilon} + \alpha T (\mathbf{u} \cdot \nabla p) + Q_L \quad (3)$$

1.1.4 Schubert et al., 2001

Schubert et al., 2001 presents three thermmal energy equations:

The general form of the energy equation (6.9.1) is:

$$\rho T \frac{Ds}{Dt} = \tau_{ij} \frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right) + \rho H \quad (4)$$

For an incompressible Newtonian fluid (eq. 6.9.13 + 6.9.14):

$$\rho C_v \frac{DT}{Dt} = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \quad (5)$$

with $\frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2$ the viscous dissipation function.

In terms of C_p instead of C_v (for a compressible fluid) (eq. 6.9.16):

$$\rho C_p \left(\frac{DT}{Dt} - \left(\frac{\partial T}{\partial p} \right)_s \frac{Dp}{Dt} \right) = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \quad (6)$$

with (eq. 6.9.15, 6.9.17):

$$\begin{aligned} \left(\frac{\partial T}{\partial p} \right)_s &= \frac{\alpha T}{\rho C_p} \\ \left(\frac{DT}{Dt} \right)_s &= \left(\frac{\partial T}{\partial p} \right)_s \frac{Dp}{Dt} = \frac{\alpha T}{\rho C_p} \frac{Dp}{Dt} \end{aligned}$$

1.2 Momentum equation

1.2.1 CRB

CRB: Cushman-Roisin and Beckers, 2011 per component

$$x : \rho \left(\frac{du}{dt} + f_* w - f v \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau^{xx}}{\partial x} + \frac{\partial \tau^{xy}}{\partial y} + \frac{\partial \tau^{xz}}{\partial z} \quad (7a)$$

$$y : \rho \left(\frac{dv}{dt} + f u \right) = -\frac{\partial p}{\partial y} + \frac{\tau^{xy}}{\partial x} + \frac{\partial \tau^{yy}}{\partial y} + \frac{\partial \tau^{yz}}{\partial z} \quad (7b)$$

$$z : \rho \left(\frac{dw}{dt} - f_* u \right) = -\frac{\partial p}{\partial z} - \rho g + \frac{\tau^{xz}}{\partial x} + \frac{\tau^{yz}}{\partial y} + \frac{\partial \tau^{zz}}{\partial z} \quad (7c)$$

1.2.2 Jarvis & McKenzie, 1980

jamc80, eq. 4 Jarvis and Mckenzie, 1980:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla P + \rho \mathbf{g} + \frac{\partial}{\partial x_j} \tau_{ij} \quad (8)$$

1.2.3 Gassmüller et al., 2020

Gassmüller Gassmüller et al., 2020:

$$-\nabla \cdot \tau + \nabla p = \rho \mathbf{g} \quad (9)$$

1.2.4 Schubert et al. 2001

$$\rho \frac{du_i}{dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i \quad (10)$$

In 2D, the momementum equation components are:

$$\rho \left(\frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y}, \quad (11)$$

$$\rho \left(\frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \right) = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \rho g. \quad (12)$$

Note that the assumption is made that gravity acts in the positive y-direction. Section 6.14?

1.3 Mass equation (continuity equation) (trivial)

jamc80, eq. 3 Jarvis and Mckenzie, 1980:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (13)$$

CRB Cushman-Roisin and Beckers, 2011:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0 \quad (14)$$

Gassmüller Gassmüller et al., 2020:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (15)$$

schubert et al. (2001) Schubert et al., 2001

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_1}(\rho u_i) = \frac{1}{\rho} \frac{D\rho}{Dt} + \frac{\partial u_1}{\partial x_1} = 0 \quad (16)$$

where $\frac{D}{Dt}$ represents the material derivative (??). For incompressible flow, $\frac{D\rho}{Dt} = 0$, so eq 16 reduces to:

$$\frac{\partial u_i}{\partial x_i} = \nabla \cdot \mathbf{u} = 0 \quad (17)$$

1.4 Starting equations (constitutive equations)

1.5 Initial and boundary conditions